Nafis Anwar

Software Engineer
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Education

University of South Florida — Tampa, FL
Bachelor of Science in Computer Science — *Expected Graduation: May 2027*

Relevant Coursework: Data Structures, Analysis of Algorithms, Object-Oriented Programming, Software System Development, Computer Organization, Computer Architecture, Machine Learning, Artificial Intelligence (Search & Planning), Reinforcement Learning, Robotics Simulation & Control, Probability & Statistics, Software Engineering, Operating Systems

Experience

Undergraduate Research Team Lead — Sep 2026 - Present
Synthetic Content & Human Interaction Study, USF — *Tampa, FL*

- Lead a **four-student research team** studying AI-like content and human engagement across Reddit communities; established study protocols and coordinated archive acquisition, data cleaning, and methodological reviews.
- Directed a **400-submission pre/post pilot** spanning **15,741 comments** and **10,142 reply edges**; developed sampling and conversation-chain protocols and evaluated **TF-IDF/MiniLM** baselines, documenting observational limitations.

Junior Undergraduate Researcher — Dec 2025 - Mar 2026
RARE Lab (USF) — *Tampa, FL*

- Operated and maintained the **Unitree G1 humanoid robot**; configured startup procedures, scripts, and evaluation harnesses for logging, benchmarking, and regression checks to support safe, reproducible research experiments.
- Conducted **literature reviews** of **sim-to-real transfer methods**, including physics mismatch mitigation, reward shaping, and delta-action policies; synthesized findings into technical summaries to inform experimental design.
- Analyzed research codebases, including the **ASAP framework**, to trace training pipelines, reproduce published results, and identify policy-learning and evaluation modules for humanoid locomotion benchmarks.

Resident Assistant — Aug 2025 - Present
Residential Education at USF — *Tampa, FL*

- Built **automated reporting and tracking** workflows using **Excel macros** and **ASP.NET Core**; documented **30+ conflict-resolution cases** and **70+ issue reports** while supporting **500+ residents**, streamlining reporting and follow-up.
- Organized and facilitated **10+ social and educational programs**, averaging **50+ attendees**; increased floor participation by **78% among 34 residents** through targeted outreach and community engagement.

Projects

EventHub Ticketing Platform | *Python, Flask, PostgreSQL, Pytest, Selenium* — Oct 2026 - Nov 2026

- Co-developed a modular **Flask ticketing demo** with event discovery, simulated checkout, currency conversion, inventory safeguards, purchase auditing, and restricted administrator workflows.
- Produced **C4 architecture models, architectural decision records, test traceability**, database migration checks, and documented release and rollback processes.
- Verified **72 passing local tests** and a successful **GitHub Actions** CI run with **87% application coverage**, PostgreSQL concurrency testing, and Selenium browser tests.

NOW_WHAT? Productivity PWA | *React, TypeScript, Dexie, IndexedDB* — Jul 2026 - Oct 2026

- Built and deployed a **local-first productivity PWA** integrating tasks, habits, goals, calendar planning, daily reviews, and reminders, with **Dexie/IndexedDB** storage and no server-side user-data collection.
- Engineered a **rule-based next-action recommendation engine** with context-aware prioritization and a debug panel to inspect its suggestions.
- Added **offline service-worker caching**, JSON import/export, cross-store integrity checks, and deterministic regression tests; published a beta using **Cloudflare Workers**.

Stock Analysis Visualizer | *C#, .NET WinForms* — Oct 2025 - Nov 2025

- Built a **Windows Forms application** to **visualize stock OHLC candlesticks** and volume from **Yahoo Finance CSVs** using the .NET charting library, with data binding directly to chart series.
- Added multi-CSV support (day/week/month), normalized OHLC ranges with padding, and indexed trading days to hide weekend/holiday gaps
- Implemented a refresh pipeline that updates charts without reloading files to improve responsiveness
- Wrote **parsing and validation checks** for malformed CSV rows and missing values to reduce crashes and improve reliability

RSA Full-Stack Educational Playground | *C, C#, ASP.NET Core, REST APIs, Docker* — Jul 2025 - Aug 2025

- Implemented key-generation, encryption, and signing flows using big-integer operations and modular arithmetic, and profiled different exponentiation strategies to **achieve 30% faster RSA operations than a baseline C# implementation.**
- Wrapped the native library in an **ASP.NET Core** backend via **C# P/Invoke** and exposed **REST endpoints** for key ops with structured JSON requests/responses and robust error handling.
- Containerized the service with **Docker** (build toolchain + native lib + runtime) and documented example curl/client calls so students can use the API consistently across environments.
- In a separate **C/POSIX threads parallel-hashing** systems project, achieved a **4.41x average speedup** over the course reference across six benchmarks.

MediTrends | *Python, Flask, FAISS, TensorFlow, PyTorch, Tailwind CSS* — May 2025 - Jun 2025

- Built **FAISS-powered semantic search** over **28k+ posts** with efficient batching (110 ms/query) and aggregation logic for volume, growth, and sentiment KPIs
- Implemented NLP pipelines for clustering and trend detection; generated KPI-based reports for community-level insights
- Developed **Flask API endpoints** for search and trending and a responsive Tailwind UI for analytical workflows
- Validated retrieval quality by testing **query sets, edge cases, and performance regressions**; documented evaluation notes and tuning decisions
- Designed the system to handle varied data sources cleanly by separating ingestion, embedding, indexing, and API layers
- Added Selenium regression tests for the Flask + UI integration to catch breaking changes in search/trending pages after backend updates

Technical Skills

Languages: Java, C++, Python, C#, TypeScript, JavaScript, SQL, C, HTML/CSS, Assembly
Backend / APIs: ASP.NET Core, Flask, REST APIs, PostgreSQL, SQLite, JSON, P/Invoke
Frontend / Storage: React, Vite, Dexie, IndexedDB, Tailwind CSS, PWAs, service workers
Testing / Automation: Pytest, Selenium, Ruff, unit/integration/security tests, Postman, root-cause debugging
Cloud / DevOps: Docker, GitHub Actions (CI/CD), Cloudflare Workers, AWS, Azure, Vercel, Amazon RDS
Data / AI / Systems: FAISS, TensorFlow, PyTorch, NLP, POSIX threads, Git, Linux/CLI, Arduino